

SYSTEM SCALABILITY & FUTURE SCOPE

Long-term Architecture Enhancements

1. Future Scalability Areas

- **Multilingual Analysis**: Leverage Whisper's multilingual decoding capabilities to grade student explanations in multiple languages.
- **LMS Integrations**: Package VBCUA as a standard LTI (Learning Tools Interoperability) module for integration into Learning Management Systems (Moodle, Canvas).
- **Cloud Deployment**: Containerize the system using Docker and deploy to AWS ECS/Fargate. Use AWS Aurora Serverless PostgreSQL and cache model weights in a shared Amazon EFS volume.